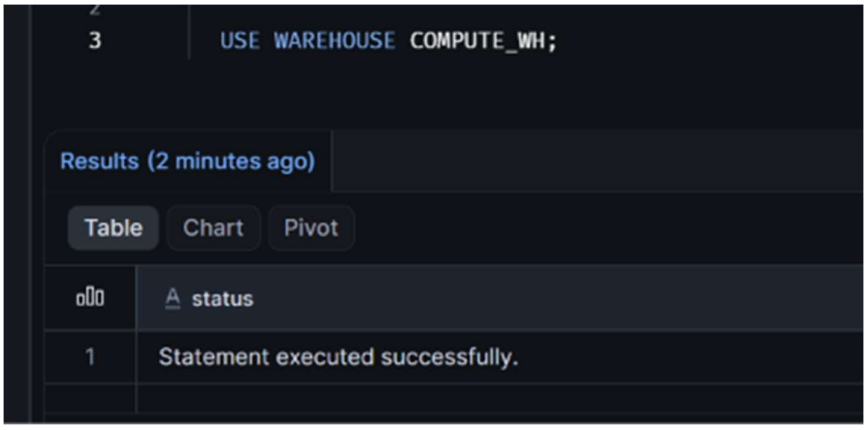
Snowflake Data Warehouse Creation and Power BI Integration

Practical Objective: To design, build, and integrate a complete cloud-to-dashboard analytics solution. This project is divided into two core phases:

- **Snowflake Infrastructure & Data Engineering:** Provisioning the necessary cloud compute architecture by configuring a virtual warehouse, creating a dedicated database and schema, and setting up file formats and stages to successfully ingest raw sample sales data into a structured table.
- **Power BI Integration & Visual Analytics:** Establishing a seamless data connection between Snowflake and Power BI Desktop to import, transform, and model the data. This culminates in the development of a professional, interactive dashboard utilizing KPI cards, date hierarchies, and diverse chart types to extract actionable business insights.

SNOWFLAKE:

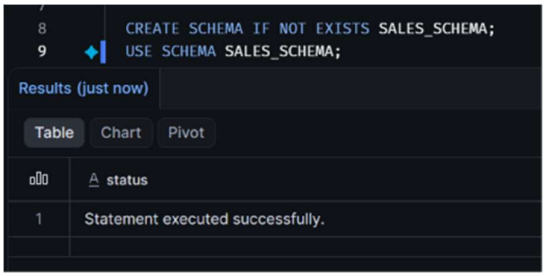
1. Configuration of Snowflake Virtual Warehouse:
   Activate the COMPUTE_WH virtual warehouse to provision the necessary computational resources for executing subsequent SQL statements.



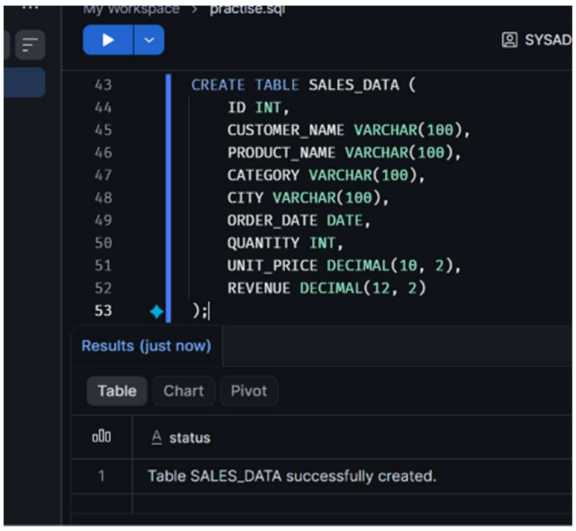
2. Database Creation:
   Create a new database named POWERBI_SALES_DB to isolate the project's data environment, and set it as the active database for the current session.



3. Schema Creation:
   Establish a schema named SALES_SCHEMA within the active database to logically organize the tables and stage objects.



4. Table Creation:
   Define the SALES_DATA table with explicitly typed columns (ID, Customer Name, Product Name, Category, City, Order Date, Quantity, Unit Price, and Revenue) to hold the structured dataset



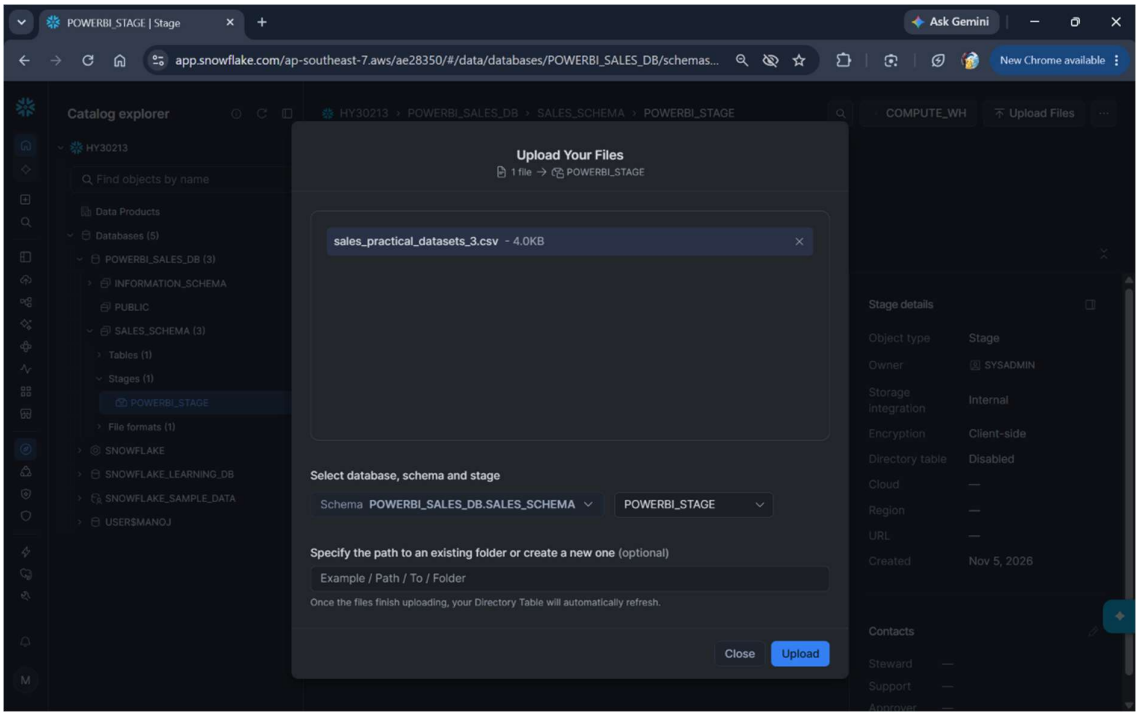
5. File Format:
   Create a file format named CSV_POWERBI_FORMAT configured to read comma-separated values and skip the initial header row during data ingestion.



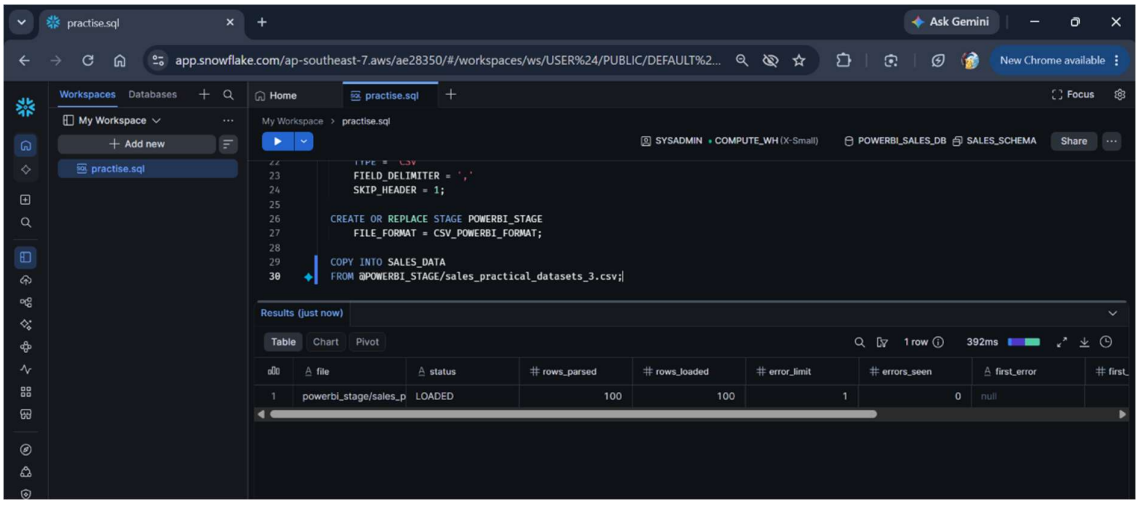
6. Stage Creation:
   Use the Snowflake Catalog Explorer web interface to manually upload the local sales_practical_datasets_3.csv file into the configured stage.



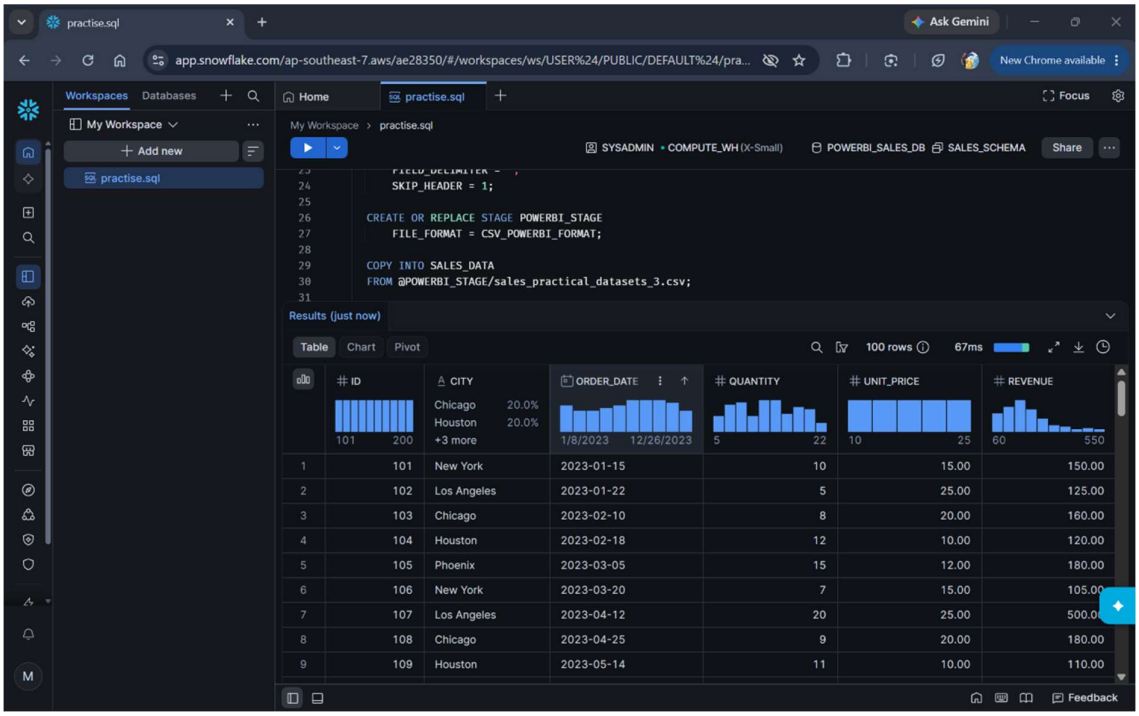
7. Sales Data Upload:
   Use the Snowflake Catalog Explorer web interface to manually upload the local sales_practical_datasets_3.csv file into the configured stage.



8. Data is Loaded to Table:
   Execute the COPY INTO SQL command to extract the data from the staged CSV file and load it permanently into the SALES_DATA table, successfully parsing the incoming rows.

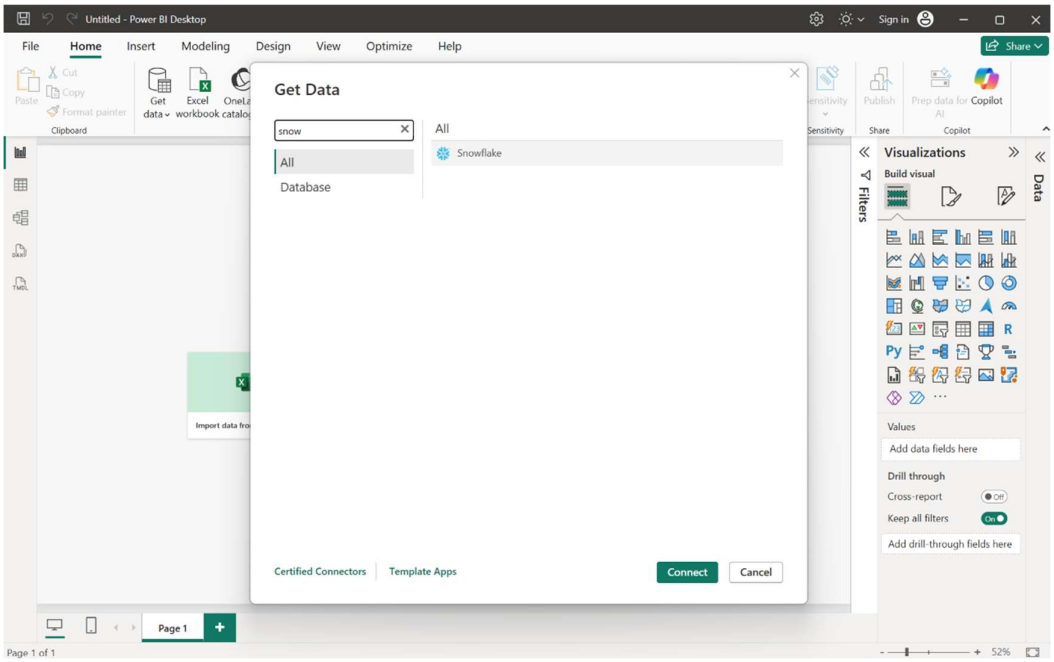


9. Verification:
   Run a SELECT * query against the SALES_DATA table to preview the dataset, ensuring all columns are properly populated and correctly formatted for downstream tools.

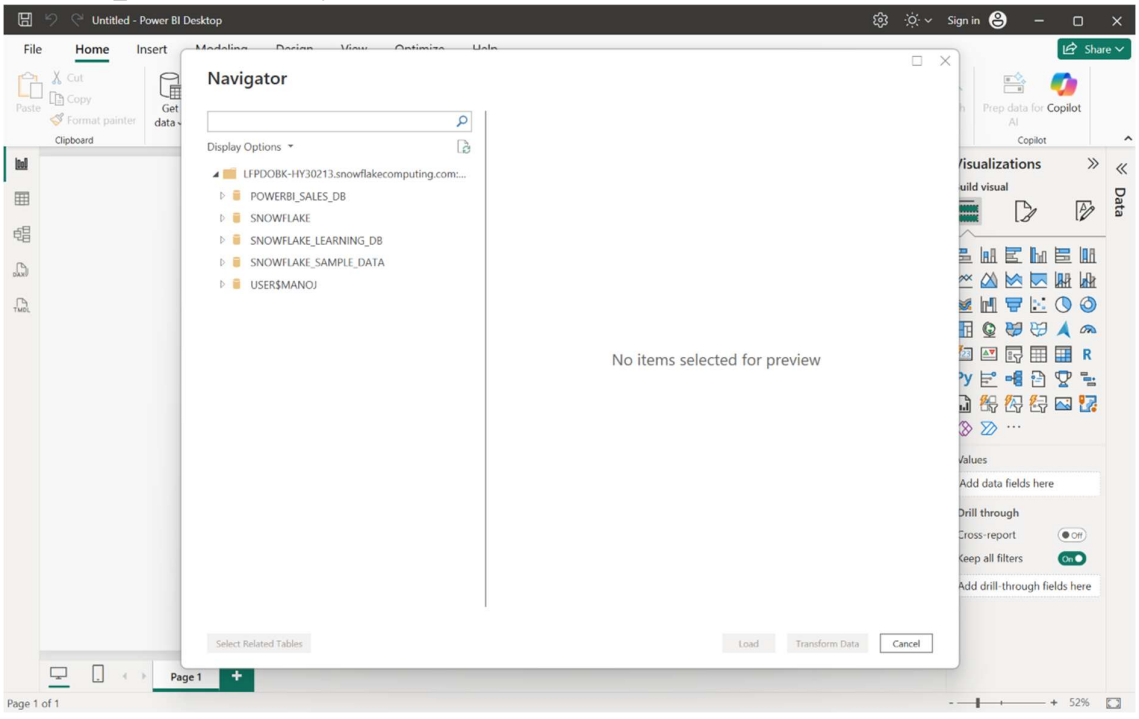


POWER BI:

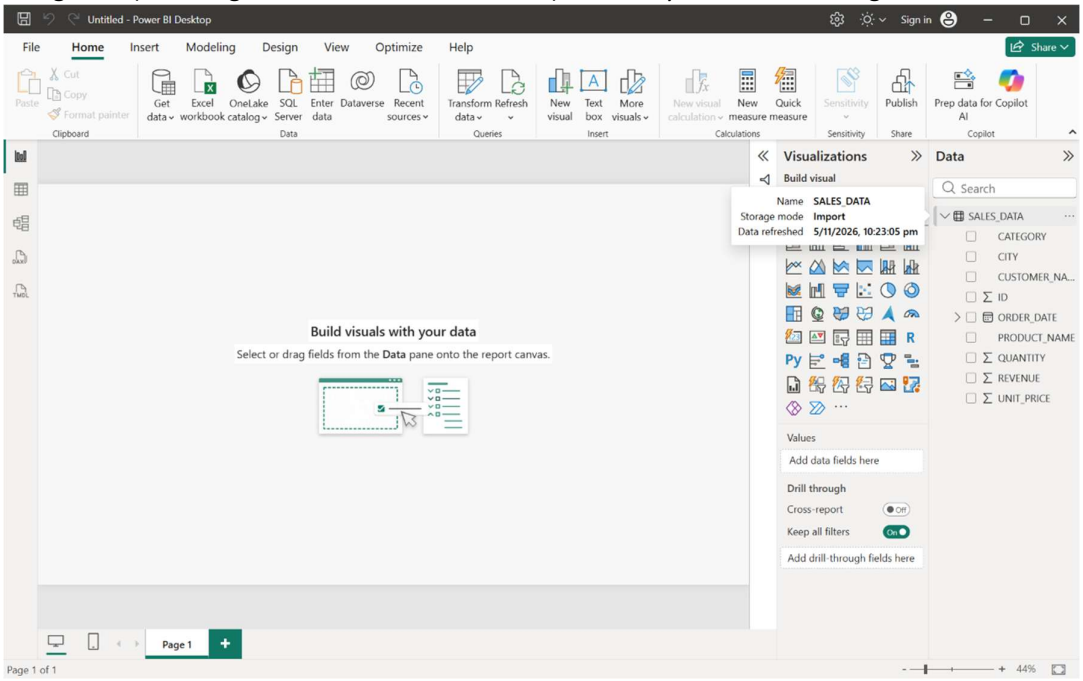
1. Snowflake Source Selection:
   Open Power BI Desktop's "Get Data" menu and select the Snowflake database connector to establish a link to the cloud data warehouse.



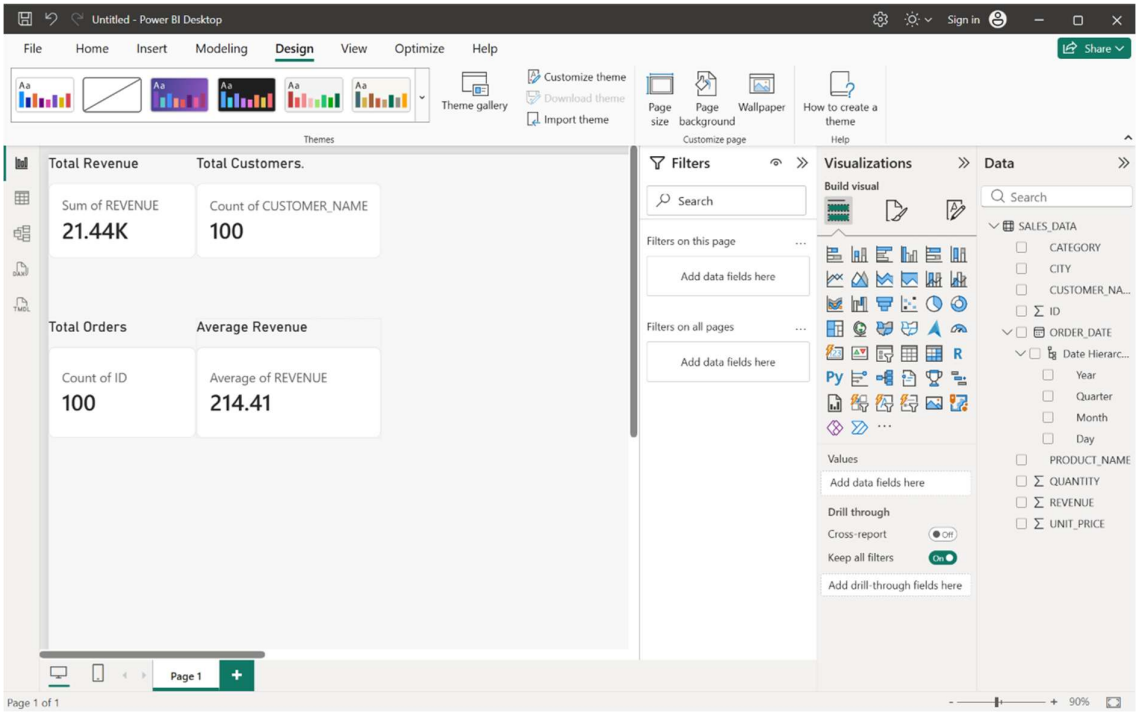
2. Navigator:
   Browse the Snowflake server hierarchy, locate the POWERBI_SALES_DB database, and select the SALES_DATA table for import.



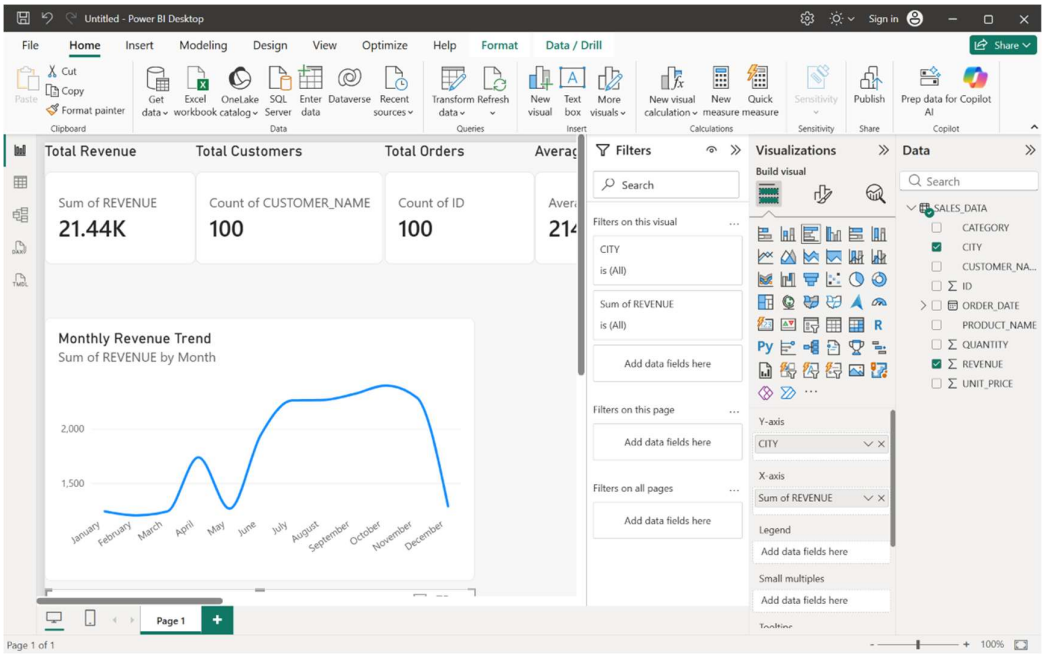
3. Sales Data is Transformed and Loaded:
   Verify that the imported SALES_DATA table appears in the Data pane with all fields properly recognized (including dates and numerical values) and ready for visual building.



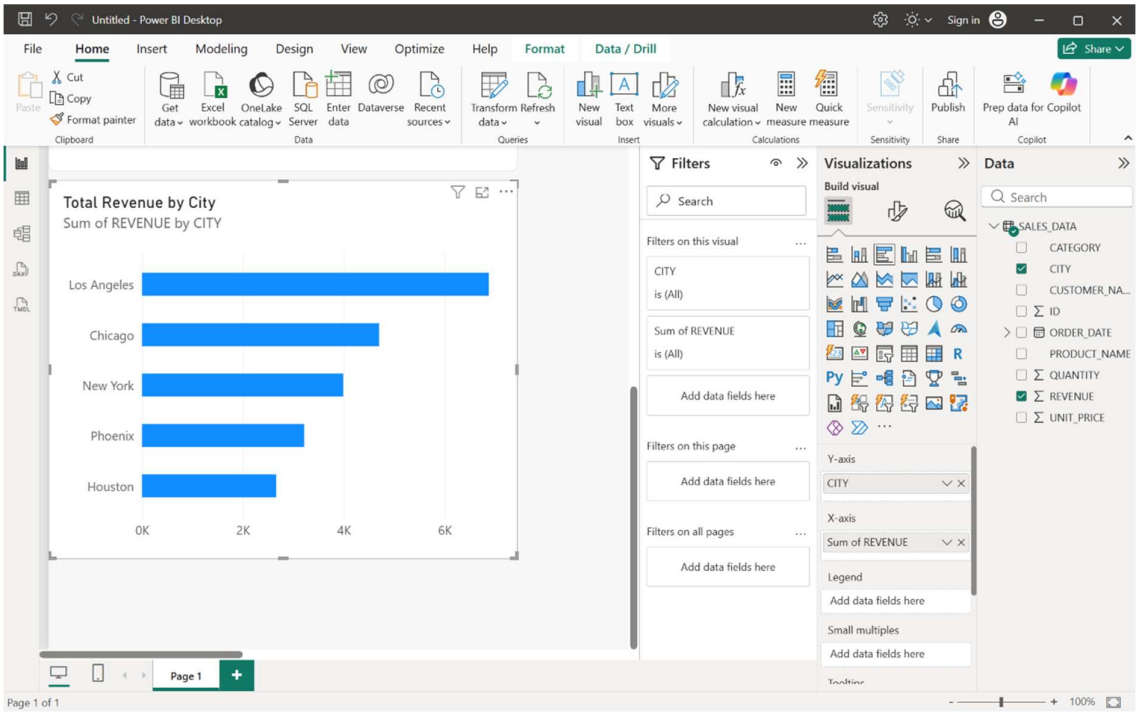
4. Initial Cards
   Create four KPI card visuals aligned horizontally at the top of the canvas to display primary aggregate metrics: Total Revenue, Total Customers, Total Orders, and Average Revenue.



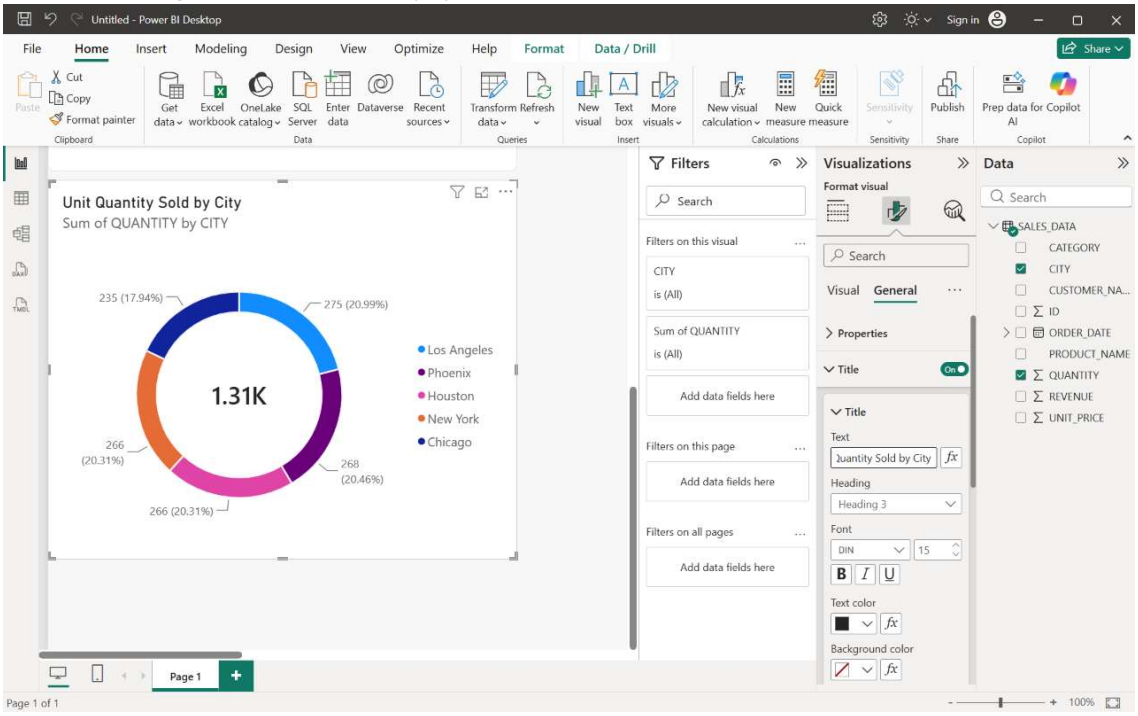
5. Line Chart – Monthly Revenue Trend:
   Plot ORDER_DATE (drilled down to the Month level) on the X-axis against the sum of REVENUE on the Y-axis to visualize revenue performance over time



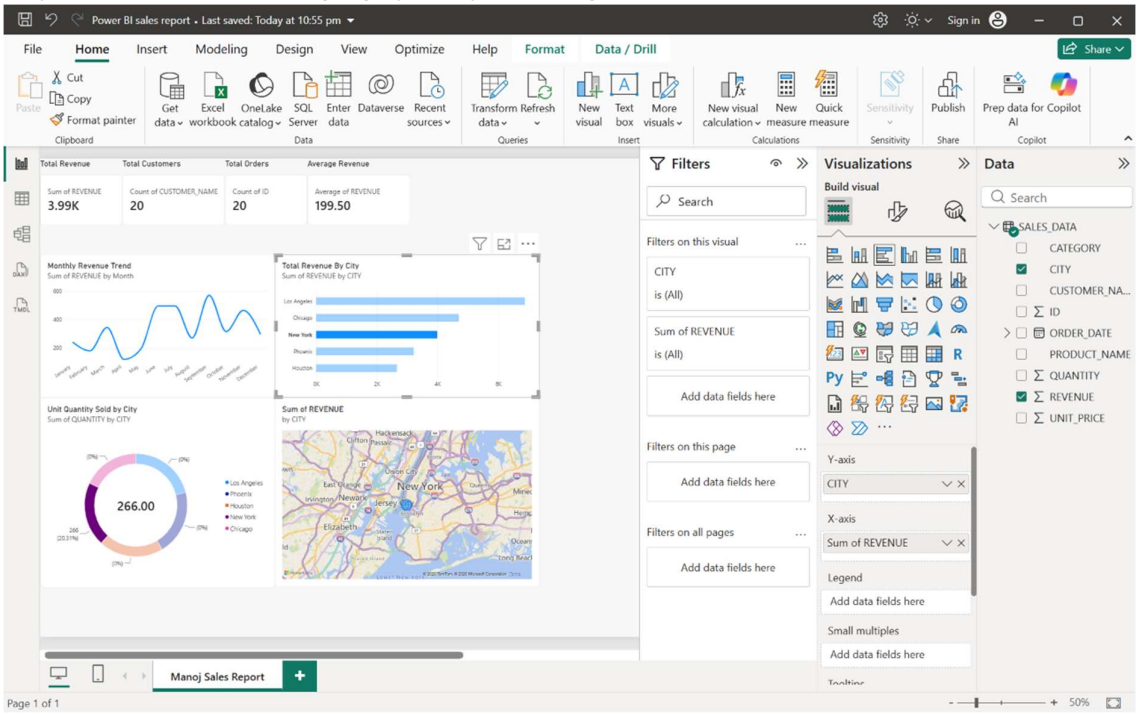
6. Bar Chart – Total Revenue By City:
   Configure a clustered bar chart with CITY on the Y-axis and the sum of REVENUE on the X-axis to compare total sales across different geographic locations.



7. Unit Quantity Sold by City:
   Design a donut chart using CITY as the legend and the sum of QUANTITY as the values to illustrate the regional distribution of physical units sold.



8. Map Visual:
   Plot CITY in the Location field and sum of REVENUE in the Bubble Size field to dynamically map revenue concentration geographically across regions.



9. Final Report:
   Assemble all visual components—including the top KPI cards, line chart, bar chart, donut chart, map, and additional visuals like the funnel and matrix—into a strictly aligned, uniformly sized grid to produce a polished, professional dashboard

